A3-PS016-TC230

Simulation-Based Handover Decision Evaluation for High-Mobility Users

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Problem Description

Home broadband users often experience slow internet speeds, packet loss, and high latency but struggle to identify the exact cause of these issues.

Key Issues:

- **Wi-Fi Problems:** Weak signals and wireless interference affect connectivity.
- **Router Issues:** Router faults can cause connection instability and high latency.
- **DNS Failures:** Slow or unresponsive DNS resolvers delay website access.
- **Packet Loss:** Lost network packets reduce connection reliability.
- **Network Congestion:** Heavy traffic increases latency and slows performance.
- **ISP Connectivity Issues:** Upstream network problems can disrupt internet access.
- **Manual Troubleshooting:** Identifying faults requires time and technical knowledge.
- **Lack of Clear Reports:** Users need understandable diagnostic results for effective troubleshooting and ISP support escalation.

Background

Background and Proposed Optimization

Background:

- In LTE and 5G networks, Event A3 triggers a handover when a neighboring cell's signal becomes stronger than the serving cell's signal by a configured offset for a specific duration.
- Short Time-to-Trigger (TTT) causes unnecessary handovers, while long TTT can delay handovers and cause connection drops.

Proposed Optimization:

- Use intelligent handover decisions based on vehicle direction, signal strength variation, historical handover patterns, and cell congestion.
- Adapt handover timing dynamically to reduce unnecessary handovers, prevent connection drops, and improve network performance.

Literature & RFC Review

- **3GPP TS 36.331:** Defines Event A3 handover triggering based on signal strength measurements. Fixed thresholds can cause unnecessary handovers or connection drops.
- **ns-3 Simulator:** Supports LTE network simulation to evaluate handover performance under different conditions.
- **Deep Reinforcement Learning (DRL):** Enables intelligent handover decisions but requires extensive training and computational resources.
- **Machine Learning Research:** Shows that signal trends, user mobility, and handover history can improve handover prediction.
- **Research Gap:** Existing methods face limitations in adaptability and computational efficiency. The proposed solution uses explainable ML to optimize handovers, reduce ping-pong effects, and improve connectivity.

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Proposed Solution

It is an intelligent, ML-based handover optimization system designed to improve connectivity in LTE/5G networks.

1. **Feature Extraction:** Analyses signal strength (RSRP/RSRQ), user mobility, signal trends, and handover history.
2. **Intelligent Handover:** Uses supervised machine learning to choose between staying connected, delaying handover, or triggering handover.
3. **Performance Evaluation:** Uses a Streamlit dashboard to visualize throughput, connection failures, and unnecessary handovers.
4. **Explainability:** Uses SHAP to explain the factors influencing handover decisions.
5. **Hardware Demonstration:** Uses a Raspberry Pi with status LEDs to demonstrate system decisions.

Optimization Proposed

Design Dimension	Conventional A3-RSRP Baseline	AIORI Optimization Proposed
Decision Features	Scalar RSRP delta, fixed offset, static hysteresis & TTT	Kinematic speed, direction, signal derivative, history, cell load
Ping-Pong Handling	Reactive post-hoc timer adjustments; prone to rapid reversals	Proactive ML classification suppresses unstable edge transitions
High-Speed Mobility	Elevated radio link failures (RLF) due to Doppler & late triggers	Trajectory trend prediction prevents sudden out-of-coverage drops
Interpretability	Rigid scalar threshold without diagnostic reasoning	SHAP feature importance & transparent decision rationale codes
Validation Rigor	Single-seed ad-hoc simulation runs with uncalibrated paths	Automated multi-seed Monte Carlo runs across 3 cells & 2 speed profiles
Telemetry & UI	Raw terminal logs and non-standardized text dumps	Interactive Streamlit telemetry suite + physical LED status demo

Solution Architecture & Design

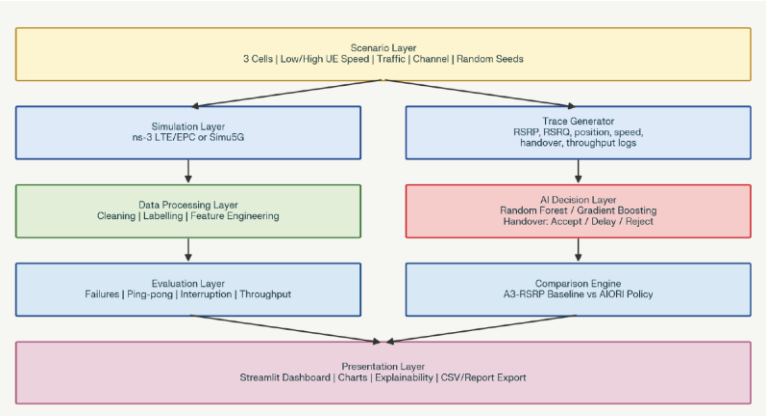


Figure: System Architecture Diagram

Data Flow Diagram

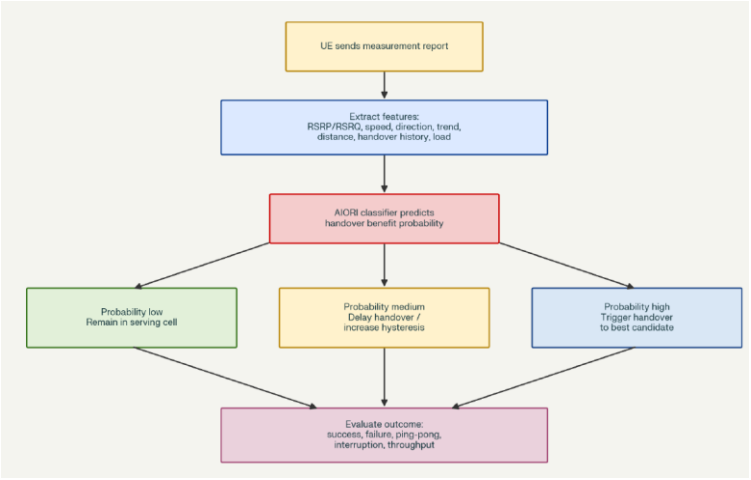


Figure: Data Flow Diagram (DFD)

Deployment Timeline

Days 1–6: Environment Setup – Configure the ns-3 LTE simulation, create network scenarios, implement baseline handover, and collect data.
Days 7–12: ML Development – Process simulation data, extract features, train a Random Forest model, and integrate it into the handover controller.
Days 13–16: Testing and Dashboard – Evaluate performance at different user speeds, measure handover failures and ping-pong events, and develop a Streamlit dashboard.
Days 17–20: Finalization and Demo – Implement SHAP explainability, integrate Raspberry Pi LEDs, prepare documentation, and rehearse the final presentation.

Methodology: Follow an iterative approach of simulation, data collection, model training, integration, performance evaluation, and final demonstration.

Bibliography I

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Thank You for Your Attention!